

HEARDS - Data Validation & Correlation

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HEARDS - Dataset - Status Update 2 Overview

- **Isolated “No Trade” rows** into a separate CSV to focus modeling on active transactions; main dataset now contains ~156,000 rows.
- **Standardized price column** by converting differentials into absolute values using corresponding MOPS benchmark prices.
- **Conducted exploratory analysis to identify anomalies and ensure dataset integrity:**
 - Verified valid date ranges, logical volume values, and consistent price_basis entries.
 - Resolved format issues and inconsistencies after merging recovered and cleaned datasets.
- **Validated data quality using DeepSeek-based review:**
 - Sampled 300 rows and re-predicted key fields (e.g., start_date, end_date, action_type, party, price, volume, location).
 - Computed accuracy and Wilson 95% confidence intervals for each field.
 - Achieved >95% accuracy across nearly all columns, confirming structural reliability.
- **Aggregated sentiment signals weekly** (Raise, Lower, Bid, Offer) and merged them with MOPS benchmark prices.
- **Standardized and resampled time series** to compute **lagged correlations** between sentiment indicators and both OTC and benchmark price movements.

No_Trade Action type

Row 1

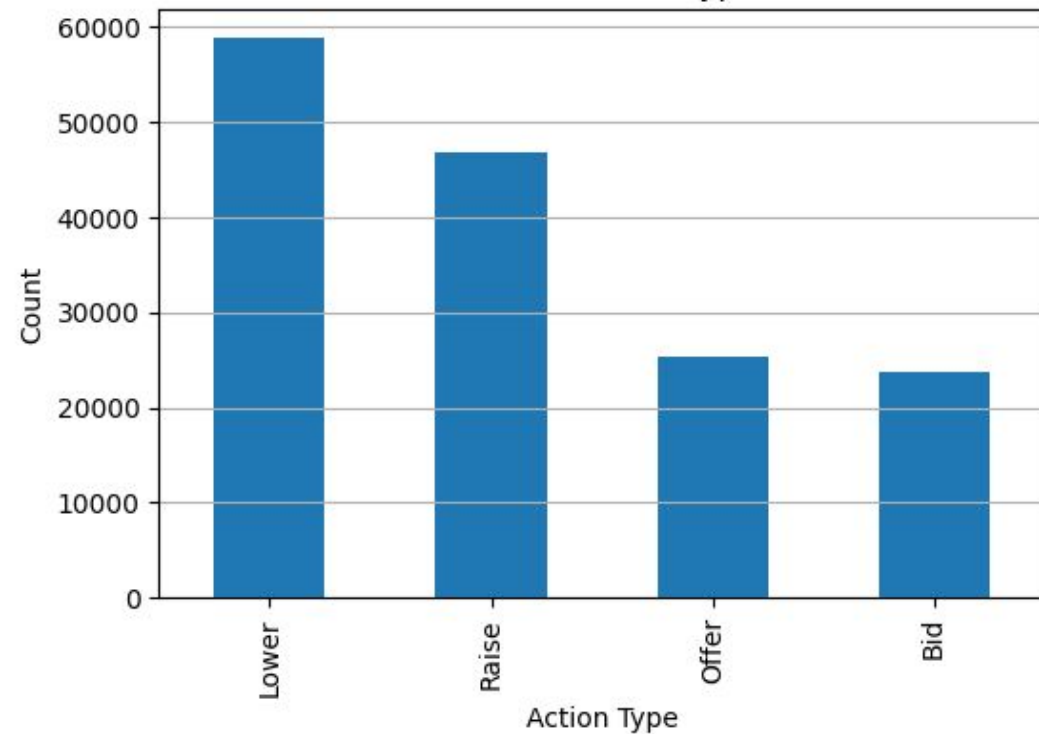
- **Party:** BRIGHTOILSG
- **Grade:** 380
- **Price:** 355.27
- **Start–End Date:** 2019-11-29 to 2019-12-25
- **Headline:**
Asia 212: Platts HSFO 380CST: Nov 29 - Dec 3 pegged at 355.27 USD/MT – No Trade

Row 2

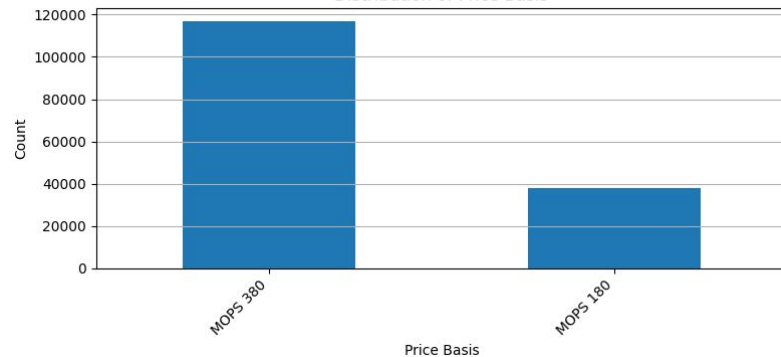
- **Party:** TRAFI
- **Grade:** 180
- **Price:** 340.20
- **Start–End Date:** 2020-10-28 to 2020-11-07
- **Headline:**
Platts HSFO 180CST: Nov 10 - Nov 14 pegged at 340.20 USD/MT – No Trade

Cleaned Dataset - EDA

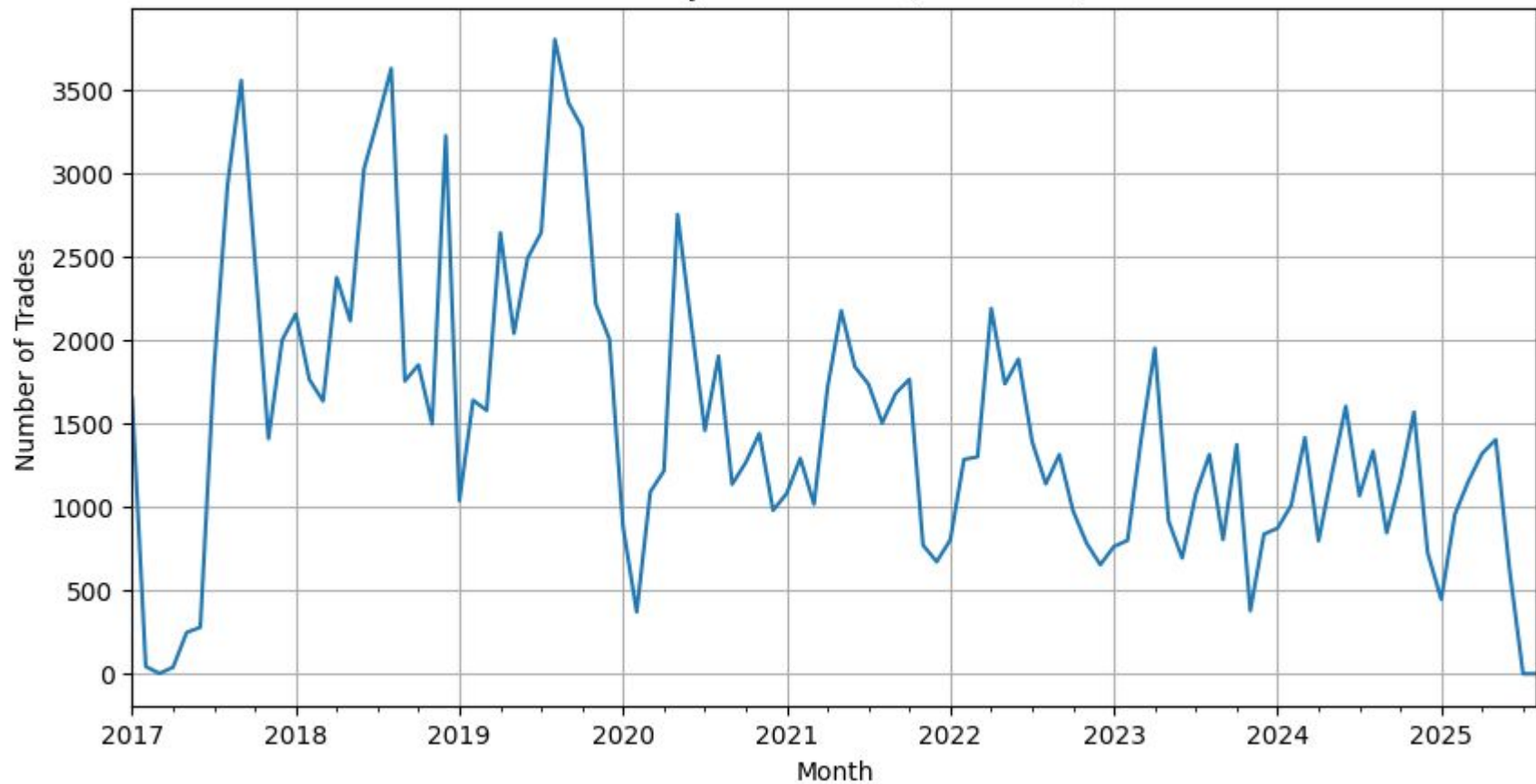
Count of Action Types



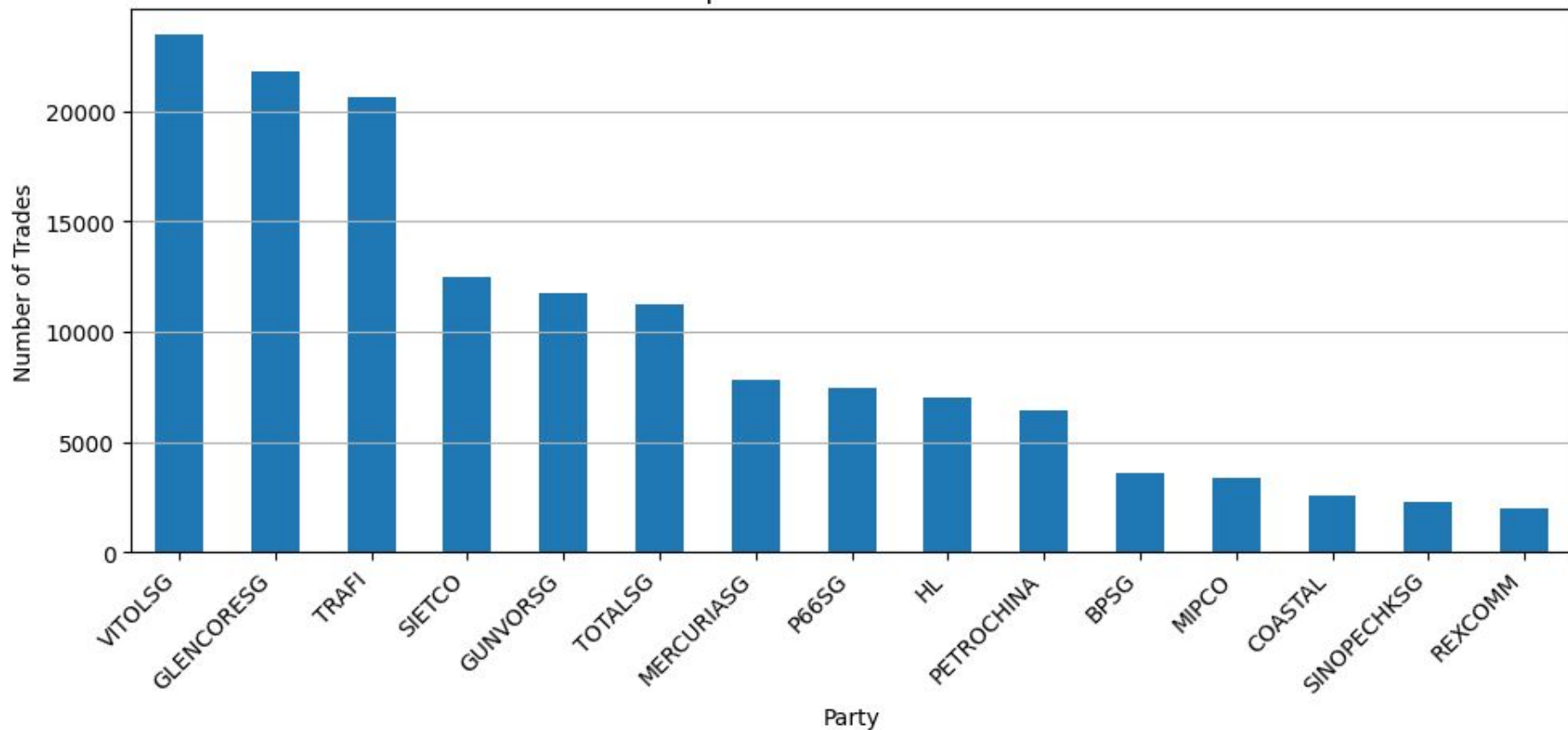
Distribution of Price Basis



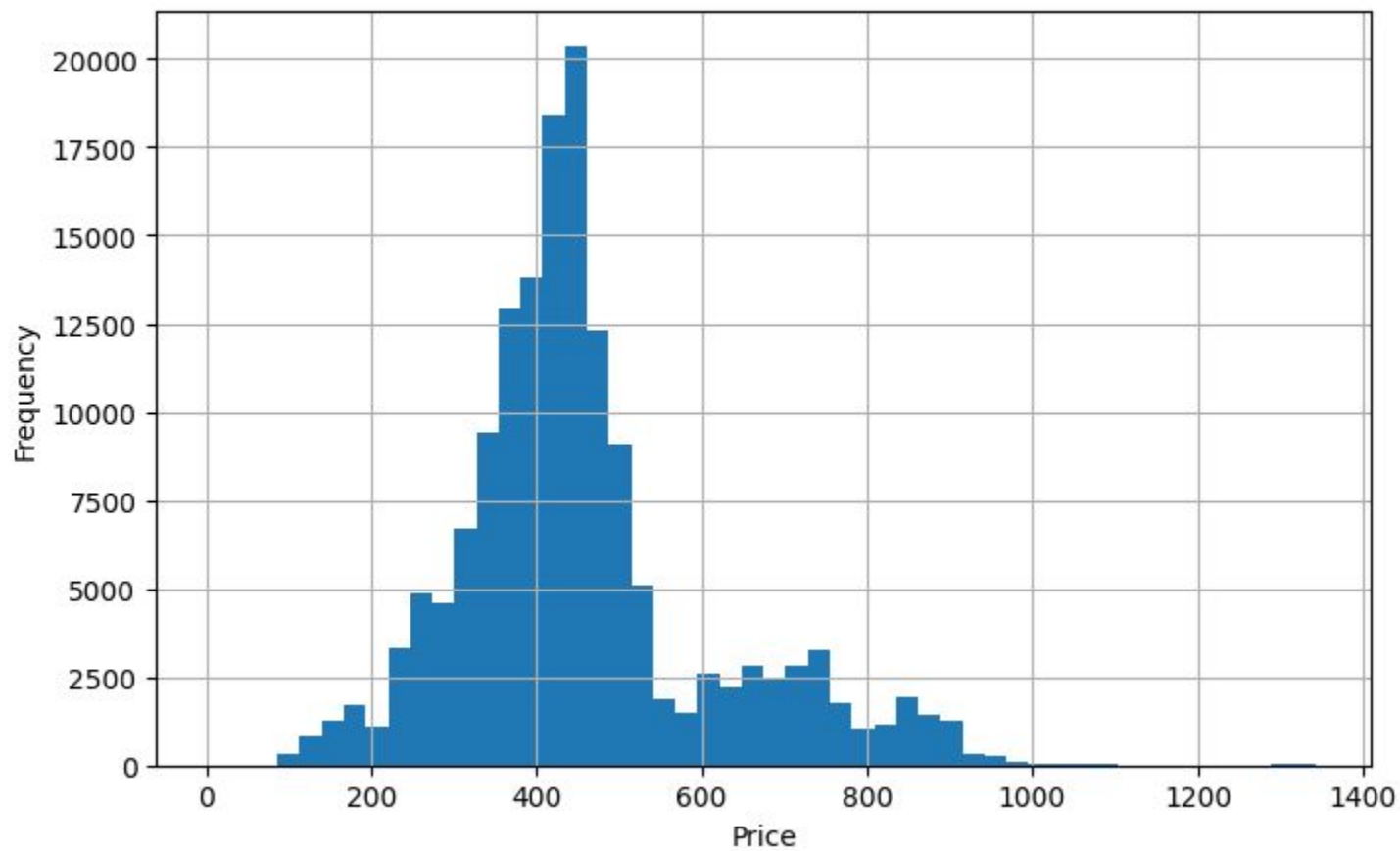
Monthly Trade Count (Start Date)



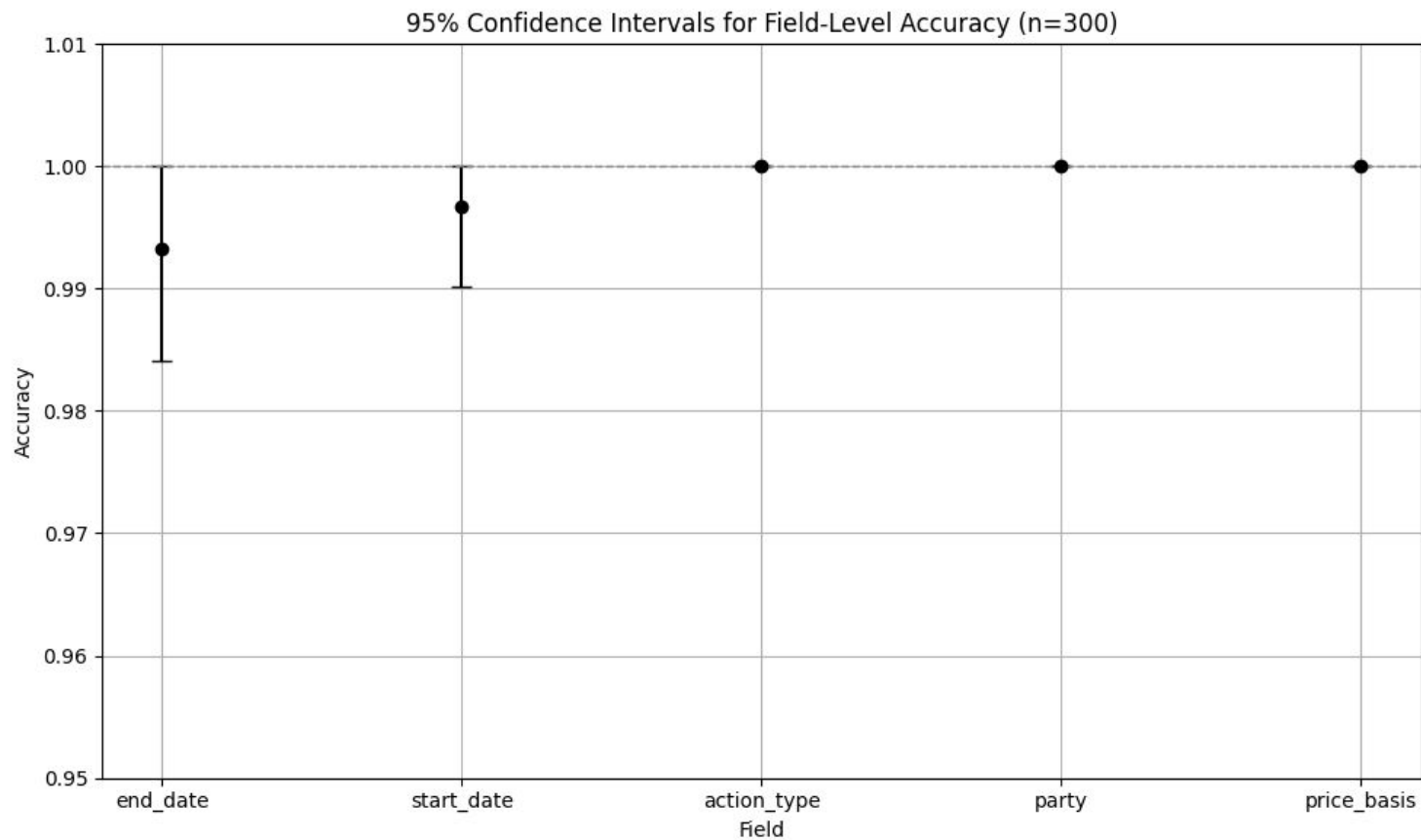
Top 15 Most Active Parties



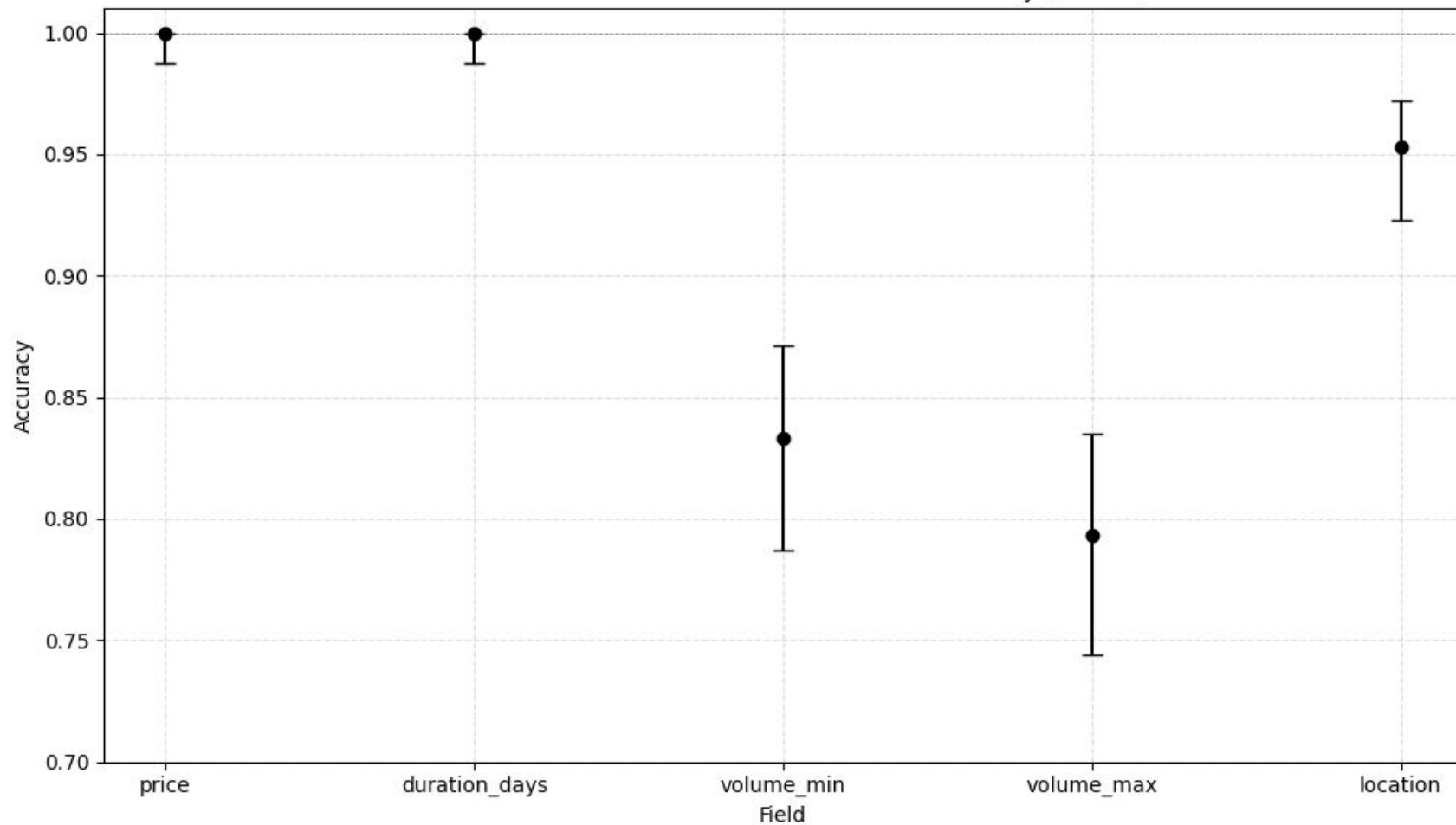
Distribution of Price



Data Validation



95% Confidence Intervals for Field-Level Accuracy (n=300)

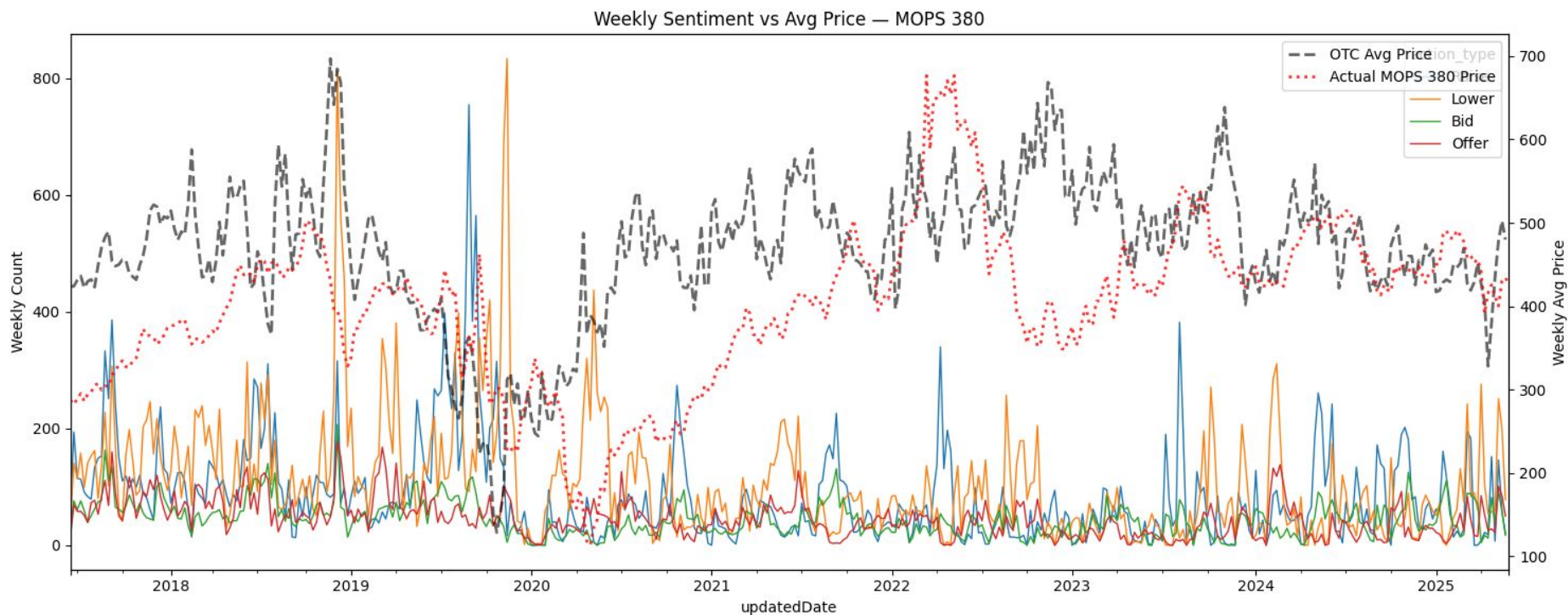


Correlational Analysis

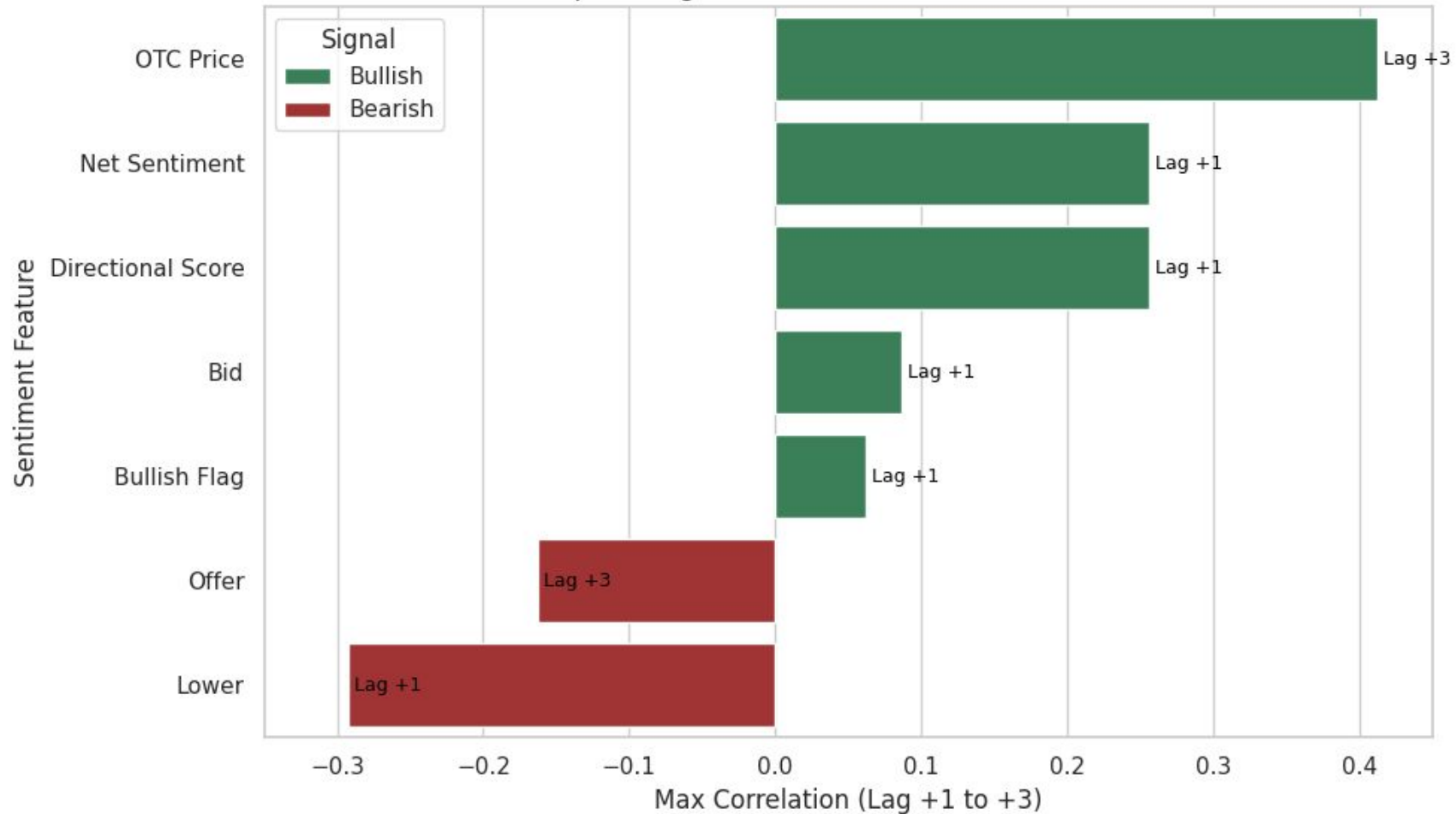
Feature Definitions and Formulas

Feature	Formula / Description
Raise	Count of HEARDS entries with action_type = 'Raise' (price increase signal)
Lower	Count of HEARDS entries with action_type = 'Lower' (price drop signal)
Bid	Count of HEARDS entries with action_type = 'Bid' (buy-side interest)
Offer	Count of HEARDS entries with action_type = 'Offer' (sell-side quotations)
Net Sentiment	Weekly total: (Raise + Bid) - (Lower + Offer), using raw counts
Directional Score	Same as Net Sentiment, but each component is z-scored before summing
Bullish Flag	Binary indicator: 1 if $z(\text{Raise}) > 1$, $z(\text{Bid}) > 1$, and $z(\text{Offer}) < -0.5$; else 0
OTC Price	Weekly average of OTC-reported transaction prices

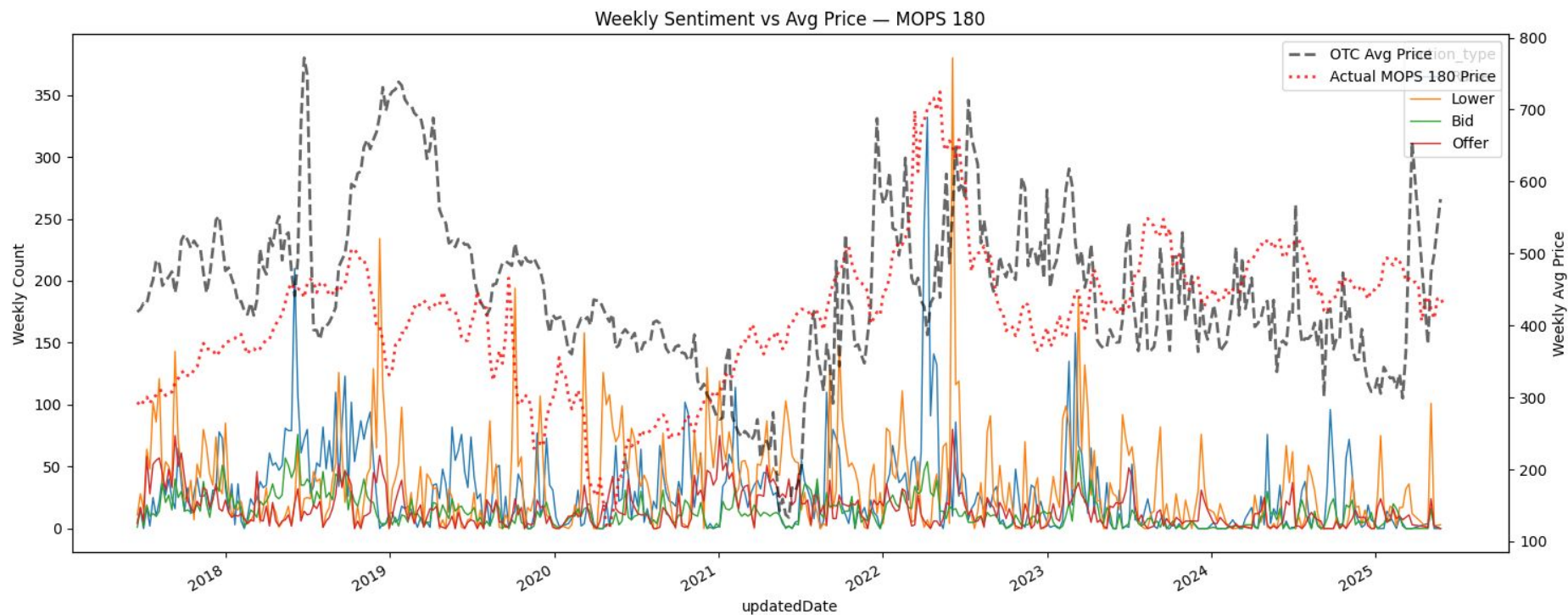
Correlational Analysis - MOPS 380



Top Leading Predictors of Future MOPS 380 Price



Correlational Analysis - MOPS 180



Top Leading Predictors of Future MOPS 180 Price

